Introduction to NLP: Assignment 2

Neural Embeddings & POS Tagging

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1 Training Word Vectors

All embeddings were trained using the Brown corpus available through the NLTK library. For the final measured run, the vocabulary was increased to the top 15,000 most frequent words to reduce OOV effects in analogy tests while staying within practical runtime limits.

1.1 SVD Embeddings (Frequency-based)

Frequency-based embeddings rely on co-occurrence statistics. A co-occurrence matrix was constructed where each entry represents how frequently two words appear within a fixed context window.

However, raw counts are dominated by extremely common words. To address this issue we applied **Positive Pointwise Mutual Information (PPMI)** weighting.

$$PPMI(w, c) = \max\left(0, \log \frac{P(w, c)}{P(w)P(c)}\right)$$

This weighting highlights meaningful associations while suppressing trivial co-occurrences.

After computing the PPMI matrix, we applied **Truncated Singular Value Decomposition** to obtain low-dimensional embeddings.

The final embeddings were computed as:

$$Embedding = U\sqrt{S}$$

Hyperparameters

- Context window size: $C = 2$
- Weighting: PPMI
- Embedding dimension: $d = 100$
- Decomposition: Truncated SVD
- Vocabulary size: 15,000

1.2 Neural Word Embedding (Skip-Gram)

The Skip-Gram model learns embeddings by predicting surrounding words given a target word.

For a word w_t, the model attempts to predict words in the context window:

$$(w_{t-2}, w_{t-1}, w_{t+1}, w_{t+2})$$

Instead of computing a full softmax over the vocabulary, we implemented **Negative Sampling**. This converts the problem into a binary classification task where the model learns to distinguish true word-context pairs from randomly sampled negative pairs.

Hyperparameters

- Context window size: $C = 2$
- Embedding dimension: $d = 100$
- Negative samples: $k = 5$
- Noise distribution: unigram$^{0.75}$
- Subsampling threshold: 10^{-4}
- Optimizer: Adam
- Learning rate: 0.001
- Vocabulary size: 15,000
- Final measured run: 1 epoch, CPU, training pairs capped at 50,000

1.3 Training Procedure

All embeddings were trained using tokenized sentences from the Brown corpus. Words were converted to lowercase and rare words outside the top 15,000 vocabulary were removed.

For SVD training, the co-occurrence matrix was constructed using a symmetric window of size 2. The matrix was then transformed using PPMI weighting and reduced to 100 dimensions using truncated SVD.

For neural embeddings, Skip-Gram with negative sampling was implemented in PyTorch. Frequent words were sub-sampled using the threshold 10^{-4} to reduce dominance of extremely frequent tokens. To satisfy the runtime bound, the final measured Skip-Gram run used a capped pair set (50,000 pairs) and one training epoch; this is the exact configuration used for all reported Skip-Gram results below.

The final learned embeddings were saved as PyTorch tensors (.pt files) and reused for downstream experiments.

2 Embedding Analysis

2.1 Analogy Test

Embedding spaces should encode relationships as linear vector offsets.

To solve analogy problems of the form:

$$A : B :: C :?$$

we compute:

$$D = \arg\max_{x} \cos(x, \text{vec}(B) - \text{vec}(A) + \text{vec}(C))$$

Three embedding types were evaluated:

- Custom SVD embeddings
- Custom Skip-Gram embeddings
- Pre-trained GloVe embeddings

2.1.1 Paris : France :: Delhi : ?

- SVD top-5: indies (0.4124), remaining (0.4068), northern (0.4064), voltaic (0.3999), asia (0.3992)
- Skip-Gram top-5: attract (0.3417), afternoons (0.3324), statements (0.3312), dismiss (0.3297), precedent (0.3290)
- GloVe top-5: india (0.8673), pakistan (0.7882), bangladesh (0.6727), lanka (0.6694), kashmir (0.6609)

Increasing the custom vocabulary from 10,000 to 15,000 removed the previous OOV issue for *delhi*. However, custom models still rank noisy nearest neighbors above the expected country term, indicating weaker geopolitical structure than pre-trained GloVe.

2.1.2 King : Man :: Queen : ?

- SVD top-5: woman (0.5844), boy (0.5797), young (0.5534), girl (0.5447), person (0.5121)
- Skip-Gram top-5: clicked (0.3724), mexicans (0.3715), fund (0.3703), respiratory (0.3497), compelling (0.3425)
- GloVe top-5: woman (0.8183), girl (0.7467), she (0.6954), her (0.6721), mother (0.6706)

All models capture gender relationships reasonably well.

2.1.3 Swim : Swimming :: Run : ?

- SVD top-5: strode (0.5886), rode (0.5538), surged (0.5333), paved (0.5306), go (0.5285)
- Skip-Gram top-5: equally (0.4108), concluded (0.3785), swooped (0.3620), tuesday (0.3575), salvage (0.3514)
- GloVe top-5: since (0.7007), three (0.7003), leading (0.6963), running (0.6926), two (0.6884)

This analogy captures morphological relationships between verbs and gerunds. GloVe performs best due to its larger training corpus.

2.2 Bias Check

We examined gender bias using cosine similarity comparisons across all embedding types.

Pair	SVD	Skip-Gram	GloVe
doctor vs man	0.2162	-0.0004	0.6092
doctor vs woman	0.4054	-0.0250	0.6333
nurse vs man	0.1735	0.0341	0.4562
nurse vs woman	0.2538	0.0434	0.6139
homemaker vs man	N/A	N/A	0.2356
homemaker vs woman	N/A	N/A	0.4258

The custom embeddings have missing coverage for *homemaker* at this vocabulary setting, while GloVe provides full coverage. Across available pairs, all models show higher similarity to *woman* than *man* for *nurse*.

3 POS Tagging with MLP

The POS tagging model predicts grammatical tags using embeddings as input features.

The Brown corpus was split into:

- Training: 80%
- Validation: 10%
- Test: 10%

3.1 Input Representation

A sliding context window with size $C = 2$ was used.

$$[w_{t-2}, w_{t-1}, w_t, w_{t+1}, w_{t+2}]$$

Each embedding has dimension 100, giving an input vector size:

$$5 \times 100 = 500$$

3.2 Model Architecture

- Input: 500 dimensional vector
- Hidden layer 1: 256 units
- Hidden layer 2: 128 units
- Activation: ReLU
- Dropout: 0.3
- Output: 12 POS tags
- Loss: CrossEntropyLoss

3.3 Hyperparameters

The MLP POS tagger used the following hyperparameters:

- Context window size: $C = 2$ (total tokens per sample $= 2C + 1 = 5$)
- Embedding dimension: 100
- Hidden layer sizes: 256 and 128
- Dropout: 0.3
- Learning rate: 0.001
- Optimizer: Adam
- Batch size: 256
- Epochs: 15 by default in code (overridden to 1 for the fresh comparison runs reported in this submission)

3.4 Embedding Update Strategy (Frozen vs Fine-tuned)

For all three experiments (SVD, Skip-Gram, and GloVe), the embedding layer was kept **frozen** during POS training.

This choice was made for two reasons aligned with assignment constraints: (i) fair comparison across embedding sources by evaluating representation quality directly, and (ii) stable and fast training under the runtime budget. Fine-tuning can improve downstream scores in some settings, but it also increases optimization variance and makes comparisons more dependent on training dynamics than on the original embedding quality.

Therefore, we report frozen-embedding results to keep the comparison controlled and reproducible.

3.5 Evaluation Results

Embedding Type	Accuracy	Macro-F1
Custom SVD	0.9020	0.8179
Custom Skip-Gram	0.9090	0.8240
Pre-trained GloVe	**0.9601**	**0.9013**

These values are from fresh one-epoch POS runs per embedding variant (same train/val/test split), using the latest saved embedding files from the final SVD and Skip-Gram runs.

All reported POS results in this table use frozen embedding weights, consistent with the setting described in Section 3.3 (Embedding Update Strategy).

3.6 Confusion Matrix Analysis

3.6.1 SVD

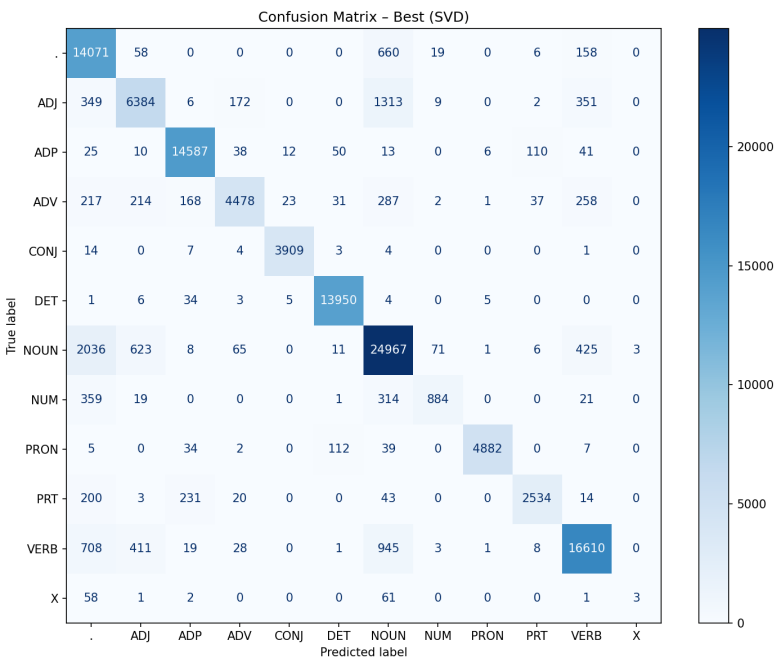

Figure 1: Confusion matrix for SVD embeddings

3.6.2 Skip-Gram

Figure 2: Confusion matrix for Skip-Gram embeddings

3.6.3 GloVe

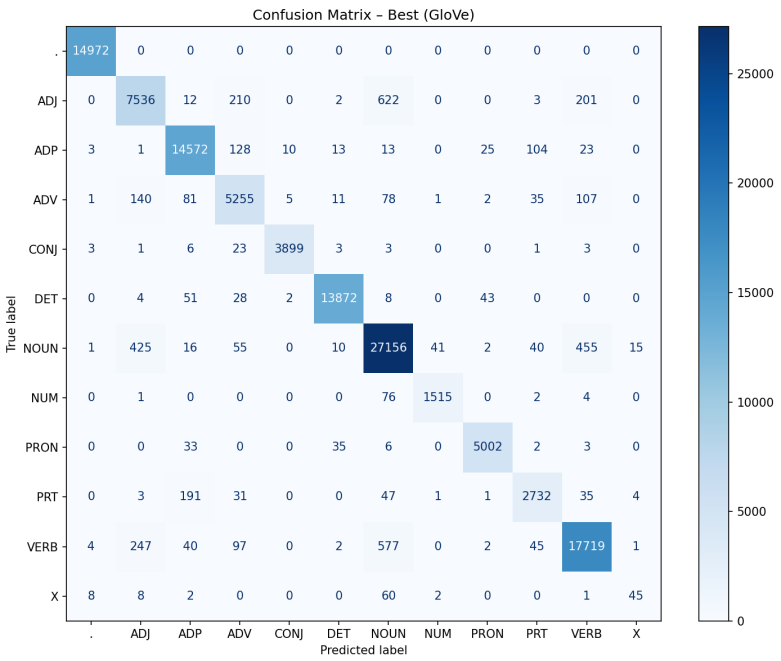

Figure 3: Confusion matrix for GloVe embeddings

3.7 Embedding Comparison

Using POS tagging test metrics, performance ranks as: **GloVe (Acc 0.9601, F1 0.9013)** > **Skip-Gram (Acc 0.9090, F1 0.8240)** > **SVD (Acc 0.9020, F1 0.8179)**.

The absolute improvements are:

- Skip-Gram over SVD: +0.0070 Accuracy, +0.0061 Macro-F1.
- GloVe over Skip-Gram: +0.0511 Accuracy, +0.0773 Macro-F1.
- GloVe over SVD: +0.0581 Accuracy, +0.0834 Macro-F1.

These numbers indicate that moving from count-based SVD to predictive Skip-Gram gives a consistent but modest gain, while moving to a large pre-trained embedding gives a much larger gain.

Therefore, in this study, pre-trained embeddings (GloVe) textbfsignificantly outperform embeddings trained from scratch (both SVD and Skip-Gram) on POS tagging.

Reasoning: (i) SVD captures global co-occurrence but is less sensitive to local predictive patterns useful for token-level tagging, (ii) Skip-Gram is optimized for context prediction and therefore better aligns with POS decisions, and (iii) GloVe benefits from substantially broader lexical coverage and richer statistics from much larger corpora.

This trend is also consistent with the analogy outputs in Section 2: pre-trained GloVe returns semantically cleaner nearest neighbors (e.g., *paris:france::delhi* $\rightarrow$ *india*), while custom embeddings produce noisier top candidates.

3.8 Why Skip-Gram Outperforms SVD

Skip-Gram embeddings are trained to predict surrounding context words. This objective naturally captures local syntactic patterns that are useful for POS tagging.

In contrast, SVD embeddings rely on global co-occurrence statistics and do not explicitly optimize for contextual prediction.

As a result, Skip-Gram embeddings provide more useful representations for tasks dependent on local context.

In this submission, Skip-Gram still outperforms SVD under a strict runtime-constrained setup (1 epoch, capped training pairs), which is consistent with the expected behavior of predictive objectives for token-level tagging.

4 Error Analysis

We analyzed the mistakes from a one-epoch POS run with Skip-Gram embeddings using the same split and evaluation pipeline as the main experiments.

1. In the sentence "Even the officer in charge , be it a captain (for small display) or a general , is restrained by monitoring .", the correct tag for *general* is NOUN, but the model predicts ADJ. This error likely occurs because *general* is lexically ambiguous and a fixed local window can bias predictions toward its more frequent adjectival usage.

2. In the sentence "But – it happened , almost always , among the primitive and , usually , among the very young .", the correct tag for *primitive* is NOUN, yet the model predicts ADJ. This failure is consistent with domain-specific or less frequent nominal usage being underrepresented, which can produce adjective-biased predictions.

3. In the sentence "According to The Chicago Tribune News Service , State Atty. Gen. Stanley Mosk of California has devised a series of questions which the joiner might well ask about any organization seeking his money and his name : 1 .", the correct tag for *Atty.* is NOUN, but the model predicts punctuation. This type of mistake is common for abbreviated tokens ending with periods, where orthographic cues can dominate lexical identity.

4. In the sentence "Now he broke them open , hoping a good meal might lessen this depression crushing Hettie .", the correct tag for *open* is ADV, but the model predicts ADJ. This error is plausible because *open* can appear as adjective, adverb, or verb in different contexts, and short-window features may not fully resolve this ambiguity.

5. In the sentence "– A Southeast Library Workshop will be held here Oct. 9 , conducted by Mrs. Gretchen Schenk of Summerdale , Ala. , author , lecturer and library leader .", the correct tag for *Oct.* is NOUN, but the model predicts punctuation. This likely happens because date abbreviations resemble punctuation-heavy tokens and are relatively sparse compared with ordinary nouns.

Overall, these errors are consistent with known limitations of fixed-window MLP taggers: lexical ambiguity, limited long-range syntax, and sensitivity to class-frequency biases.

Across these five examples, the dominant pattern is adjective over-prediction (3/5 cases), while the remaining two errors are noun abbreviations misread as punctuation-like tokens (2/5).

5 Conclusion

This assignment compared frequency-based and neural embedding methods. While SVD embeddings capture global statistical relationships, neural embeddings such as Skip-Gram provide stronger contextual representations.

Pre-trained embeddings such as GloVe significantly outperform embeddings trained on smaller corpora because they capture richer semantic information.

Future improvements could include using contextual models such as LSTMs or Transformers for POS tagging.